8.3 Rewriting Quadratic Equations

Lesson Introduction

Benchmarks	MA.912.AR.3.6 Given an expression or equation representing a quadratic function, determine the vertex and zeros and interpret them in terms of a real-world context. MA.912.AR.3.7 Given a table, equation, or written description of a quadratic function, graph that function and determine and interpret its key features. MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.	Learning Targets	• I can rewrite a quadratic equation in factored or vertex form to standard form. • I can rewrite a quadratic equation from standard form to factored form. • I can determine key features of a quadratic function from a variety of forms.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions

- Why is knowing how to write quadratics in different forms helpful in finding key features?
- How can I rewrite a quadratic equation into different forms?

Lesson Narrative

In this lesson, students explore three forms of quadratic equations: vertex form, factored form, and standard form. Students will discover which specific key features are revealed for each of these specific forms. They will practice rewriting quadratic equations into different forms in order to determine additional key features.

Component Summary

8.3.1 Warm-Up (~5 min.)	Which quadratic function graph does not belong? *(SE p. 17)*	8.3.4 Guided Instruction (10 min.)	Formal introduction to rewriting quadratic equations in factored form *(SE p. 20)*
8.3.2 Guided Instruction (10 min.)	Introduction to the three different forms of quadratics *(SE p. 18)*	8.3.5 Guided Practice (10 min.)	Identifying key features given a quadratic equation, rewriting in additional forms when needed *(SE p. 21)*
8.3.3 Collaboration (10-15 min.)	Rewriting quadratic equations in standard form collaboration activity *(SE p. 20)*	8.3.6 Wrap-Up (~5 min.)	Lesson summarizing activity *(SE p. 23)*

Resources

Glossary	Materials	Additional Preparation
<u>Key Terms:</u> • Vertex Form • Standard Form • Factored Form <u>Additional Terms</u>: N/A	• (Optional) Chart Paper • (Optional) Markers	• N/A

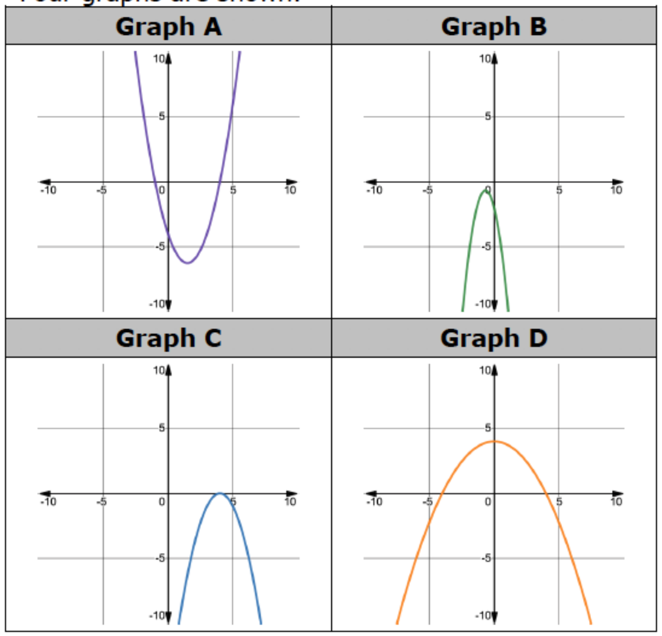
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may incorrectly multiply binomials. For example, $(x + 2)(x + 3) = x^2 + 6$.
Students may believe different forms represent different quadratics.</td><td>Consider activating prior knowledge regarding area model or distributive property when students are multiplying binomials.
Students will get more practice writing, rewriting, and graphing quadratics in Lesson 4. Prompt students to persevere through activities and errors and not to get stuck in this lesson, as they will get more practice in Lesson 4.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
8.3.3 Collaboration allows students to work with a partner to rewrite the quadratic equations into standard form. This instructional routine provides structure that positions both students as sense makers rather than just one student engaging with most of the cognitive demand.</td><td>MA.K12.MTR.1.1: Participate in Effortful Learning
Students will be working collaboratively as they learn to rewrite quadratic equations into standard form. They will need to help and support each other as they learn a new method that may be necessary to identify a specific key feature.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider spiraling content and vocabulary from Lesson 2 with student-created vocabulary strategies. As students continue to make sense of the vocabulary of key features of quadratic functions, consider which strategies might best serve the learning needs of your students.
Interactive Word Walls – give working definitions to add to or describe for auditory and written entry points
Foldables – offer physical layout options for spatial or kinesthetic entry points
Frayer Models – provide examples and nonexamples for visual entry points</td></tr>
<tr><td colspan="3">Lesson Notes</td></tr>
</table>

8.3.1 Warm-Up: Which One Doesn't Belong?

Component Overview: *Students reflect on which graph does not belong and justify their selection(s). This routine has a low-floor, as all students can participate, and high-ceiling, as students can be challenged to think of multiple reasons for each graph.*

8.3.1 Warm-Up: Which One Doesn't Belong?

1. Four graphs are shown.



   a. Which graph doesn't belong?
   Answers will vary. Any graph could be used.

   b. Justify your reasoning using key features of the graphs.
   Answers will vary. Examples include:
   Graph A is the only graph where the right and left end behavior are positive infinity.
   Graph B is the only graph with no x-intercept(s).
   Graph C is the only graph with only one x-intercept.
   Graph D is the only graph with a positive y-intercept.

Have a few students share out their answers to the Warm-Up with the class, asking them to provide a rationale or justification for their answer. Use talk moves to have students agree or disagree with the reasonableness of each answer.

8.3.2 Guided Instruction: Forms of Quadratic Equations

Component Overview: *Students will receive guided instruction on the three different forms of quadratic equations. For this activity, begin by teaching the forms of quadratic equations, using the narrative language at the beginning of this lesson. Walk students through each table, highlighting each of the forms and the key features they reveal. Prompt students to answer question 1 verbally through the instructional delivery. For question 2, choose two lines (such as 1 and 4) to model the thinking for this table. Then call on students to try the other ones to help guide thinking.*

8.3.2 Guided Instruction: Forms of Quadratic Equations

Quadratic equations can be represented in three forms: vertex form, factored (x-intercept) form, and standard form.

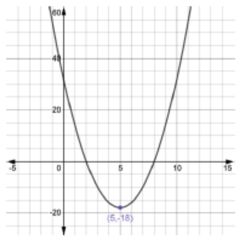
Forms of Quadratic Equations

Vertex form of a quadratic equation is represented as $y = a(x-h)^2 + k$, where (h, k) is the vertex.

Factored form of a quadratic equation is represented as $y = a(x-p)(x-q)$, where $(p, 0)$ and $(q, 0)$ are the x-intercepts.

Standard form of a quadratic equation is represented as $y = ax^2 + bx + c$, where $(0, c)$ is the y-intercept.

Each form highlights key features that can be used to assist in graphing quadratic equations.

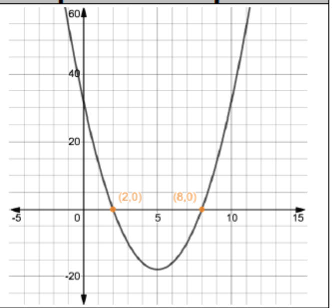
Form of the Equation	Graph of the Equation
Vertex form of a quadratic equation $y = a(x-h)^2 + k$ Example shown: $y = 2(x-5)^2 - 18$ Vertex $(5, -18)$ Axis of symmetry $x = 5$ Parabola opens up	
Key Features	

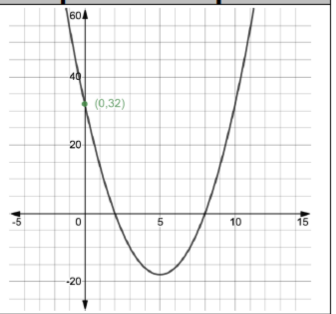
• Highlights the vertex of a parabola (h, k) and the axis of symmetry $x = h$.
• Highlights whether the graph opens up or down.
  ○ If $a > 0$ (positive), the graph opens up.
  ○ If $a < 0$ (negative), the graph opens down.	

Consider leveraging an Anchor Chart or other classroom display that showcases each of the three forms of quadratic equations. Leave space for "key features revealed" that you can go back to later as the lesson progresses.

How does each form show a different key feature?

8.3.2 Guided Instruction: Forms of Quadratic Equations (continued)

Form of the Equation	Graph of the Equation
Factored form of a quadratic equation $y = a(x - p)(x - q)$ Example shown: $y = 2(x - 8)(x - 2)$ x-intercepts $(8,0)(2,0)$ Parabola opens up	
Key Features	
• Highlights the intercepts of a parabola $(p,0)(q,0)$. • Highlights whether the graph opens up or down.	

Form of the Equation	Graph of the Equation
Standard form of a quadratic equation $y = ax^2 + bx + c$ Example shown: $y = 2x^2 - 20x + 32$ y-intercept $(0, 32)$ Parabola opens up	
Key Features	
• Highlights the y-intercept $(0, c)$. • Easily shows the coefficients for a, b and c, which can be used to determine other key features of quadratics. • Highlights whether the graph opens up or down.	

Utilize talk moves to leverage key features for each form as an instructional tool for learning. Prompt students to make sense of the key features in their own words and reference examples.

If you created the Anchor Chart referenced in the earlier portion of this activity, it is important that the information from the Key Features section of the table be transferred to the Anchor Chart.

1. What do you notice about the graphs of the three different quadratic equations?
Answers will vary. All of the graphs are the same even through the equations are written differently.

2. Complete the table for each equation.

Equation	Form of the Equation	Key Feature(s) Revealed	Value(s) of Key Features
$y=(x-8)(x+13)$	○ vertex form ○ *factored form* ○ standard form	*x-intercepts* *Direction of parabola*	$(8,0)$ *and* $(-13,0)$ *Opens Up*
$y=-\frac{1}{2}x^2-2x-2$	○ vertex form ○ factored form ○ *standard form*	*y-intercept* *Direction of parabola*	$(0,-2)$ *Opens Down*
$y=(x+6)^2-4$	○ *vertex form* ○ factored form ○ standard form	*Vertex* *Axis of Symmetry* *Direction of parabola*	$(-6,-4)$ $x=-6$ *Opens Up*
$y=-x^2-12x+36$	○ vertex form ○ factored form ○ *standard form*	*y-intercept* *Direction of parabola*	$(0,36)$ *Opens Down*
$y=3(x+4)^2$	○ *vertex form* ○ factored form ○ standard form	*Vertex* *x-intercept* *Axis of Symmetry* *Direction of parabola*	$(-4,0)$ $x=-4$ *Opens Up*

How does your understanding of equivalence help support that a parabola can be represented by different forms? Summarize the key features that were easily found from each form.

For question 2, show students how to complete the table using two lines from the table for an "I Do" before releasing. Be sure not to choose two lines, such as 2 and 4, that have the same form of the equation so that students get to practice with multiple forms of the equation. Consider using line 1 as an "I Do" and line 4 as a "We Do" and release the remaining for students to try.

- *How did you determine what form the quadratic equation is in?*
- *How did you determine which key features would be revealed?*

Some students may mark the last row as factored form of a perfect square trinomial from the work they did in Unit 7. These students may choose other key features that can be determined from this form. Point out to students that noting this connection will be helpful in Lesson 4. Assure students that either classification is correct. Help students who classify y = 3(x+4)2 as a factored form to notice that it is also vertex form where the y-value of the vertex is zero. Student responses may vary and grow as they become more familiar with the different forms. Allow students to share their answers and their thought processes so all students can benefit from the different connections each makes as they develop understanding.

For an additional challenge, have students create an equation of their own. They can take each of their creations and trade with another student to have them determine which form of the equation it is in and what key features are revealed.

8.3.3 Collaboration: Rewriting Quadratic Equations – Part 1

Component Overview: *Students will work together to rewrite quadratic equations into standard form. For this activity, release students to work independently on questions 1 and 2 and then together on 3 and 4. Call the class back together to review answers as a whole group.*

8.3.3 Collaboration: Rewriting Quadratic Equations - Part 1

1. Rewrite the quadratic equation in standard form.

$$y = (x-3)(x+1)$$

$y = x^2 + x - 3x - 3$
$y = x^2 - 2x - 3$

Activate prior knowledge regarding distributive property or the area model for students to avoid incorrectly multiplying the binomials.

2. Rewrite the quadratic equation in standard form.

$$y = (x-1)^2 - 4$$

$y = (x-1)(x-1) - 4$
$y = x^2 - x - x + 1 - 4$
$y = x^2 - 2x - 3$

3. With your partner, compare the two standard form equations and write what you observe.
Answers will vary. Both standard forms are the same. Both equations, after re-writing were the same when in standard form.

Questions for consideration:
- *What form is question 1 written in?*
- *What key features does question 1 reveal?*
- *What form is question 2 written in?*
- *What key features does question 2 reveal?*

4. Discuss with your partner if any key features could help determine if a quadratic in vertex form and a quadratic equation in factored form are equivalent. Write a conjecture.
Answers will vary. Both forms reveal the direction the quadratic is facing. Calculating the vertex could help determine if they are equivalent.

During the review of this activity, allow multiple student pairs to share their conjectures with their rationale that support their idea.

8.3.4 Guided Instruction: Rewriting Quadratic Equations – Part 2

Component Overview: *Students will receive guided instruction on how to rewrite quadratic equations into factored form, building from what they learned in Unit 7 from factoring trinomials.*

8.3.4 Guided Instruction: Rewriting Quadratic Equations - Part 2

1. Rewrite the quadratic equation $y = -x^2 - 4x + 5$ in factored form and identify the x-intercepts.
   $y = -(x + 5)(x - 1)$
   x-intercepts: $(-5,0)$ *and* $(1,0)$

2. Liam wants to jump over 15 chairs on his dirt bike. To calculate where the landing ramp must be placed, in meters, he uses the quadratic equation $y = -3x^2 + 21x$. Rewrite the equation in factored form and identify the x-intercepts.
   $y = -3x(x - 7)$
   x-intercepts: $(0,0)$ *and* $(7,0)$

Consider adding to or creating a new Anchor Chart, note-taking guide, or Foldable that illustrates the connections between the different forms of a quadratic equation. Students can also use the classroom Anchor Chart and make one of their own during this time.

3. A ball is thrown straight up from 2 meters above the ground with a velocity of 7 meters/second. The height of the ball is represented by the equation $y = -4x^2 + 7x + 2$, where x is time in seconds.

   a. Rewrite the equation in factored form.
      $y = -1(4x + 1)(x - 2)$

   b. Discuss with your partner what key features can be determined from each of the equations.

Equation	**Standard Form** $y = -4x^2 + 7x + 2$	**Factored Form** $y = -1(4x + 1)(x - 2)$
Key Features Revealed	*y-intercept:* $(0,2)$ *Direction of Parabola: opened downwards*	*x-intercepts:* $(-\frac{1}{4}, 0)$ *and* $(2,0)$ *Direction of Parabola: opened downwards*

For questions 1-3
- *For each context, what key features may be useful to know?*
- *Would knowing end behavior be useful in a real-world context? Give an example.*

8.3.4 Guided Instruction: Rewriting Quadratic Equations – Part 2 (continued)

c. Identify the x-intercepts.

$\left(-\frac{1}{4},0\right)$ *and* $(2,0)$

d. Discuss with your partner what the x-intercepts mean in terms the given context. Complete the sentence.

Because x is
- ○ time, in seconds,
- ○ height, in meters,

the x-intercepts give the
- ○ time
- ○ height

when the ball will
- ○ reach the ground.
- ○ be at its greatest height.

How can you determine what the x-intercepts mean in the context? What key feature represents the alternative? (Maximum height, or y-value of the vertex represents, represents the ball at its highest point.)

8.3.5 Guided Practice: Rewriting Quadratic Equations and Key Features

Component Overview: *Students will practice identifying key features given a quadratic equation, rewriting into additional forms when needed.*

8.3.5 Guided Practice: Rewriting Quadratic Equations and Key Features

1. For each given equation, rewrite it into the specified form. Then, use both equations to determine key features of the quadratic. Show your work.
   It may not be possible to determine all key features from the given form and the rewritten form.

Factored Form		Standard Form	
$y = (x+7)(x-6)$		$y = x^2 + x - 42$	
Key Features			
x-intercept	$(-7,0)$	increasing	$-0.5 < x < \infty$
x-intercept	$(6,0)$	decreasing	$-\infty < x < -0.5$
y-intercept	$(0,-42)$	positive	$x < -7$ *or* $x > 6$
vertex	$(-0.5,-42.25)$	negative	$-7 < x < 6$
domain	$\mathbb{R}$	opens up/down	*Up*
range	$-42.25 \le y < \infty$	axis of symmetry	$x = -0.5$
left end behavior	$x \to -\infty$, $y \to \infty$	right end behavior	$x \to \infty$, $y \to \infty$

Factored Form		Standard Form	
$y = -(x+15)(x-2)$		$y = -x^2 - 13x + 30$	
Key Features			
x-intercept	$(-15,0)$	increasing	$-\infty < x < -6.5$
x-intercept	$(2,0)$	decreasing	$-6.5 < x < \infty$
y-intercept	$(0,30)$	positive	$-15 < x < 2$
vertex	$(-6.5,72.25)$	negative	$x < -15$ *or* $x > 2$
domain	$\mathbb{R}$	opens up/down	*Down*
range	$-\infty < y \le 72.25$	axis of symmetry	$x = -6.5$
left end behavior	$x \to -\infty$, $y \to -\infty$	right end behavior	$x \to \infty$, $y \to -\infty$

Consider your students when engaging with this Guided Practice.

- *Pull a small group for intervention while other students practice independently or in pairs.*
- *Have students attempt the first two on their own before clarifying misconceptions whole group, then release students to finish the final two.*
- *Have students pair up or jigsaw questions 1 and 2 before trying 3 and 4 on their own.*

What are the benefits of using the factored form when finding key features?

8.3.5 Guided Practice: Rewriting Quadratic Equations and Key Features (continued)

Vertex Form		Standard Form	
$y = -(x-3)^2 + 1$		$y = -x^2 + 6x - 8$	
Key Features			
x-intercept	$(2,0)$	increasing	$x < 3$
x-intercept	$(4,0)$	decreasing	$x > 3$
y-intercept	$(0,-8)$	positive	$2 < x < 4$
vertex	$(3,1)$	negative	$x < 2$ *or* $x > 4$
domain	$\mathbb{R}$	opens up/down	*Down*
range	$y \leq 1$	axis of symmetry	$x = 3$
left end behavior	$x \to -\infty$, $y \to -\infty$	right end behavior	$x \to \infty$, $y \to -\infty$

Standard Form		Factored Form	
$y = 2x^2 + 14x + 12$		$y = 2(x+6)(x+1)$	
Key Features			
x-intercept	$(-6,0)$	increasing	$-3.5 < x < \infty$
x-intercept	$(-1,0)$	decreasing	$-\infty < x < -3.5$
y-intercept	$(0,12)$	positive	$x < -6$ *or* $x > -1$
vertex	$(-3.5,-12.5)$	negative	$-6 < x < -1$
domain	$\mathbb{R}$	opens up/down	*Up*
range	$-12.5 \leq y < \infty$	axis of symmetry	$x = -3.5$
left end behavior	$x \to -\infty$, $y \to \infty$	right end behavior	$x \to \infty$, $y \to \infty$

Consider using a "Rapid Whiteboard Exchange" using vocabulary when reviewing the answers. For example, provide the numeric answer and have students hold up the correct key feature that it represents. For example, write the point $(0,-8)$. Students with the correct answer will hold up a whiteboard or paper that displays "y-intercept" on it.

What are the benefits of using the standard form when identifying key features?

If you use the "Rapid Whiteboard Exchange," consider adding a layer of challenge to the process. Use values that are not part of the two forms and allow students to use an answer of "not applicable" or another equivalent indicator.

8.3.6 Wrap-Up: Cool Down

Component Overview: *Students will complete a summarizing activity of this lesson. This activity is designed to be done independently.*

8.3.6 Wrap-Up: Cool Down

1. Write a note to a friend who missed class today summarizing today's lesson.

   Answers will vary.

Give students specific constraints based on the information you are targeting as the teacher.

For example: "Write at least five sentences. Include at least three vocabulary words. Include one example."